Question #1 of 120

Question ID: 1207634

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Gabe Klein, CFA, an analyst for HB Investments, is responsible for the valuation model for an IPO. Without his knowledge, others at HB adjusted the inputs to the model to increase the estimated value of the shares, and the offering is oversubscribed. Complying with local securities laws, Klein purchases shares of the IPO for his personal account and allocates the remaining shares to client accounts on a pro rata basis. With regard to the Standard on knowledge of the law, the analyst:

- **A)** did not violate the Standard.
- **B)** **violated the Standard by purchasing the shares of the IPO but not by allowing the IPO valuation to be published.**
- **C)** violated the Standard by allowing the IPO valuation to be published and by purchasing the shares of the IPO.

Explanation

Standard I(A) Knowledge of the Law requires candidates and members comply with all applicable rules and regulations, including the CFA Institute Standards of Practice. Further, Standard I(A) requires that members and candidates must not knowingly participate in violations of applicable laws and Standards. Even though local law permits purchasing shares for personal accounts before purchasing IPO shares for client accounts, Standard VI(B) Priority of Transactions does not. The analyst knowingly violated the Code and Standards and, thus, violated Standard I(A). Since the analyst was unaware of the deceit in the valuation of the IPO stock [a violation of Standard I(D) Misconduct], his participation in publishing the IPO valuation did not constitute a violation. (Modules 3.1, 3.8)

For Further Reference:

(Study Session 1, Modules 3.1, 3.8, LOS 3.a, 3.b, 3.c)

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Question #2 of 120

Question ID: 1207637

Green Investments utilizes the CFA Institute Standards of Professional Conduct as their standards for ethical practice. For purposes of compliance, which of the following is *least likely* a violation of Green Investments' policies?

- **A)** **One of Green Investments' marketing brochures states that several of the firm's portfolio managers passed all three levels of the CFA exam on their first attempts.**
- **B)** At a meeting with potential clients, Green's chief investment officer states that he is among a group of the most qualified investment professionals because he holds the CFA charter.

C) In interviewing a prospective employee, a portfolio manager at the firm says that the position could be financially rewarding because CFA charterholders are known to achieve superior performance results.

Explanation

The chief financial officer and the portfolio manager are in violation of Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program because they have improperly referenced the CFA designation or exaggerated its meaning (e.g., "group of the most qualified"). The marketing brochure is only stating factual information regarding the portfolio managers' success at passing the CFA exams on their first attempts. (Module 3.9)

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

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Question #3 of 120

Question ID: 1207631

Charmaine Townsend, CFA, has been managing equity portfolios for clients using a model that identifies growth companies selling at reasonable multiples. With economic growth slowing for the foreseeable future, she has decided to change to a securities selection model that emphasizes dividend income and low valuation. To comply with the Code and Standards, Townsend should *most appropriately*:

A) promptly notify her clients of the change.

B) get written permission from her clients prior to the change.

C) get written acknowledgment of the change from her clients within a reasonable period of time after the change is made.

Explanation

Standard V(B) Communication with Clients and Prospective Clients requires prompt disclosure of any change that might significantly affect the manager's investment processes. The disclosure need not be in writing. (Module 3.7)

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

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Question #4 of 120

Question ID: 1207628

Wells Investments implements a new procedure for unsolicited trade requests that an advisor believes are inconsistent with the client's IPS:

- If the trade will have only a minimal impact on the client's portfolio, first advise the client in what way the trade deviates from the IPS, and then request the client's approval for the trade.
- If the trade will have a material impact on the risk and return characteristics of the client's portfolio, discuss with the client that this trade will require a change in the IPS.

Which of these statements is consistent with the standard concerning suitability?

A) **Both of these statements.**

B) Neither of these statements.

C) Only one of these statements.

Explanation

If an unsolicited trade is inconsistent with a client's IPS, a member or candidate should not execute the trade before discussing it with the client. According to Standard III(C) Suitability, if the trade will have only a minimal impact on the client's portfolio, the member or candidate should attempt to educate the client with regard to how it deviates from the IPS and then may follow her firm's policies for obtaining client approval for the trade. If the trade will have a material impact on the risk and return characteristics of the client's portfolio, the discussion should focus on changing the IPS. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

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Question #5 of 120

Question ID: 1207625

With respect to the Standard on material nonpublic information, materiality is *least likely* to be affected by:

A) the source of the information.

B) **liquidity of the subject security.**

C) ambiguity about the price effect of the information.

Explanation

According to Standard II(A) Material Nonpublic Information, how specific the information is, how different it is from public information, and its nature are key factors in determining whether a particular piece of information fits the definition of material. An additional factor is reliability, which is often a function of the source of the information. While the liquidity of a security may be a factor in determining the materiality of advance knowledge of a large buy or sell order, in most cases, it would not be a factor in determining materiality. (Module 3.3)

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

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Question #6 of 120

Question ID: 1207632

Alberto Cosini is the top-rated, sell-side analyst in the biotechnology industry. His recommendations significantly affect prices of industry stocks regularly. Yesterday Cosini changed his rating on Biopharm from "hold" to "buy," and Cosini's firm emailed the change to its clients although no public disclosure has yet been made. If Peter Allen, CFA, who heard about Cosini's rating change for Biopharm from his brother, purchases Biopharm in his personal account, Allen will *most likely*:

A) **not violate the Standards.**

B) violate the Standard concerning diligence and reasonable basis.

C) violate the Standard concerning material nonpublic information.

Explanation

There is no requirement that a firm publicly release ratings changes by its analysts. Individuals outside the firm acting on this information after it is released to clients are not in violation of the Standard concerning nonpublic information. Purchases in a member's personal account are not subject to the requirements of the Standard concerning diligence and reasonable basis, so there is no violation indicated here. (Modules 3.3, 3.7)

For Further Reference:

(Study Session 1, Modules 3.3, 3.7, LOS 3.a, 3.b, 3.c)

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Question #7 of 120

Question ID: 1207623

Judy Dudley, CFA, is an analyst and plans to visit a company that she is analyzing in order to prepare a research report. The Standard related to independence and objectivity:

A) requires Dudley to pay for her own transportation costs and not to accept any gifts or compensation for writing the report, but allows her to accept accommodations and meals that are not lavish.

B) requires Dudley not to accept any compensation for writing a research report, but allows her to accept company paid transportation, lodging, and meals.

C) **allows Dudley to accept transportation, lodging, expenses, and compensation for writing a research report, but requires that she disclose such an arrangement in her report.**

Explanation

Standard I(B) Independence and Objectivity allows investor-paid research but requires that members and candidates limit the type of compensation they accept for writing a research report so that it is not dependent on the conclusions of the research report. Best practice is for analysts to only accept a flat fee for such company-paid research reports. Such research should also include complete disclosure of the nature of the compensation received for writing such a report so that investors will not be misled as to the relationship between the analyst and the company. Paying for one's own transportation and lodging when the analyst is not employed by the subject firm is a recommended procedure for complying with Standard I(B), but it is not a requirement. (Module 3.1)

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

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Question #8 of 120

Question ID: 1207633

Campbell Hill, CFA, has recently accepted the position of Chief Compliance Officer at an investment management firm. Hill distributes a memo stating that effective immediately (1) material supporting all company research reports will be kept in the company database in electronic form for 10 years, and hard copies of the same material will be maintained for one year only, and (2) hard copy records of all trade confirmations sent to clients must be kept on file for five years, the period mandated by local regulations. With respect to record retention:

A) **neither of Hill's policies violates the Standards.**

B) Hill's policies regarding both research reports and trade confirmations violate the Standards.

C) Hill's policy regarding research reports does not violate the Standards, but the policy regarding trade confirmations does.

Explanation

In the absence of regulatory requirements, Standard V(C) Record Retention recommends maintaining records supporting investment recommendations and actions and records of investment-related communications with clients for at least seven years. Here, there is regulatory guidance, and seven years is a recommendation, not a requirement, in any case. Records can be maintained in electronic or hard copy format. (Module 3.7)

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

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Question #9 of 120

Question ID: 1207639

In calculating total firm assets for a GIPS-compliant performance statement, Allen Bund, CFA, finds that there is a mix of fee-paying and non-fee-paying accounts, some of which are discretionary and some of which are non-discretionary accounts. Should Bund include non-discretionary accounts and non-fee-paying accounts in the calculation of total firm assets?

	Non-discretionary accounts	Non-fee-paying accounts
A)	**Yes**	**Yes**
B)	No	Yes
C)	No	No

Explanation

Total firm assets must include all accounts whether discretionary or not and whether fee-paying or not.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.a)

CFA® Program Curriculum, Volume 1, page 235

Question #10 of 120

Question ID: 1207627

Dawn Shepard, CFA, is a broker for a regional brokerage firm. Her company's research department recently changed its recommendation on the common stock of Orlando (ORL) from "buy" to "sell" and sent the change to all firm clients who own ORL. The next day, a client places a "buy" order for ORL. According to the Standards, under these circumstances, Shepard:

A) must advise the customer of the change in recommendation before accepting the order.

B) has complied with the fair dealing Standard and may accept the order because it is unsolicited.

C) may accept the order only if the customer acknowledges in writing that she was notified of the change in the recommendation.

Explanation

Under Standard III(B) Fair Dealing, clients placing orders contrary to the firm's changed recommendation should be advised of the change in recommendation before the firm accepts the orders. (Module 3.4)

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

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Question #11 of 120

Question ID: 1207635

Paul James, CFA, a retail stock broker, notices that one client in particular, Chet Young, Ph.D., is especially adept at picking stocks. James decides to replicate Young's trades in his own account after he enters them. By doing so, James:

A) is not in violation of any Standards.

B) is in violation of the Standard on priority of transactions because he is front running the client's account.

C) is in violation of the Standard on misconduct because he has misappropriated confidential client information.

Explanation

James is not in violation of the Standards. To comply with Standard VI(B) Priority of Transactions, members and candidates must give transactions for clients and employers priority over their personal transactions. In this instance, James did not adversely affect the client's interest because the client's trades were executed before James copied them. He has not acted fraudulently or deceitfully and, thus, has not violated Standard I(D) Misconduct. (Modules 3.2, 3.8)

For Further Reference:

(Study Session 1, Modules 3.2, 3.8, LOS 3.a, 3.b, 3.c)

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Question #12 of 120

Question ID: 1207624

Nicholas Hart, CFA, is a portfolio manager for individuals. Last year, Hart's wife was hospitalized for several months. Despite his best efforts to pay her bills, Hart was forced to declare personal bankruptcy but did not disclose this to his clients. According to the CFA Institute Standards of Professional Conduct, Hart:

A) **is not in violation of any Standard.**
B) is in violation of the Standard on communication with clients for not disclosing his bankruptcy to his clients.
C) is in violation of the Standard on misconduct for personal conduct that reflects adversely on his professional reputation.

Explanation

The circumstances of Hart's bankruptcy do not compromise his professional reputation. The bankruptcy did not involve fraudulent or deceitful business conduct; therefore, there is no violation of Standard I(D) Misconduct. The Standards do not require disclosing the bankruptcy to clients because it does not create any conflict of interest and is not relevant to Hart's professional activity. (Module 3.2)

For Further Reference:

(Study Session 1, Module 3.2, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #13 of 120

Question ID: 1207636

Marie Marshall, CFA, charges clients a management fee and commissions on securities transactions. Marshall receives an annual bonus based on the overall success of the firm and a quarterly bonus based on the trading volume in her clients' accounts. If Marshall does not tell clients about her compensation package, she is violating the Standard concerning:

A) **disclosure of conflicts.**
B) communication with clients.
C) additional compensation arrangements.

Explanation

Marshall has an obligation to disclose that she receives special compensation based on the amount of client trading volume. Standard VI(A) Disclosure of Conflicts requires members to disclose to clients and prospects all matters that could potentially impair the member's ability to make investment decisions that are (and to give investment advice that is) objective and unbiased. The Standard on communications with clients addresses issues that involve clearly communicating investment recommendations and analysis. The Standard on additional compensation arrangements is concerned with accepting benefits that may create a conflict between a member's interests and her employer's interests. (Module 3.8)

For Further Reference:

(Study Session 1, Module 3.8, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #14 of 120

Question ID: 1207640

Lunar Wealth, a subsidiary of Galaxy Financial, has prepared GIPS- compliant performance data and asks Galaxy's president about his interest in presenting GIPS-compliant performance data, but he does not believe it is a priority. Lunar may:

A) claim partial compliance with GIPS if Lunar's performance presentations are in compliance.
B) not claim compliance with GIPS because compliance must be made on a company-wide basis.
C) **claim compliance with GIPS as long as Lunar is presented to the public as a distinct business entity.**

Explanation

Lunar may claim compliance as long as it has met the reporting requirements necessary and is held out to clients (advertised) as a distinct business entity. Claiming partial compliance is not allowed.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.b)

CFA® Program Curriculum, Volume 1, page 236

Question #15 of 120

Question ID: 1207629

Fred Reilly, CFA, is an investment advisor. Roger Harrison, a long-term client of Reilly, decides to move his accounts to a new firm. In his review of Harrison's account history, Reilly discovers some transfers of funds from the account of Harrison's company that Reilly suspects were illegal. Which of the following actions is *most appropriate* for Reilly to take under the Standards?

A) **Discuss his suspicions with outside counsel.**
B) Inform Harrison's company of the suspected illegal activities because Harrison is no longer a client.
C) Do nothing because he must maintain the confidentiality of client information even after the client has left the firm.

Explanation

Of the choices given, seeking the advice of outside counsel about what actions Reilly may be required to take is the most appropriate. Under Standard III(E) Preservation of Confidentiality, members and candidates should maintain the confidentiality of information received in the course of their professional service relating to both current and former clients. In the case of illegal activity, however, Reilly may have a legal obligation to report the activity or, on the other hand, may have a legal obligation to maintain the client's confidentiality even if he suspects illegal activity. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #16 of 120

Question ID: 1207626

Fred Dean, CFA, has just taken a job as trader for LPC. One of his first assignments is to execute the purchase of a block of East Street Industries. While working with East Street on an assignment for his previous employer, he learned that East Street's sales have weakened and will likely be significantly below the LPC analyst's estimate, but no public announcement of this has been made. Which of the following actions would be the *most appropriate* for Dean to take according to the Standards?

A) Contact East Street’s management and urge them to make the information public and make the trade if they refuse.

B) Request that the firm place East Street’s stock on a restricted list and decline to make any trades of the company’s stock.

C) Post the information about the drop in sales on an internet bulletin board to achieve public dissemination and inform his supervisor of the posting.

Explanation

Standard II(A) Material Nonpublic Information requires that members and candidates who possess material nonpublic information not act or cause others to act on the information. Refusing the trade would violate this Standard because it would be acting or causing others to act on the nonpublic information. Dean should seek to have East Street make the information public. If East Street does not do so, Dean must act as he would have acted if he did not possess the information. Refusing to make the trade he was instructed to make would be "acting" on the information in this case. The obligation here is to the integrity of financial markets. (Module 3.3)

For Further Reference:

(Study Session 1, Module 3.3, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

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Question #17 of 120

Question ID: 1207630

When members and candidates report performance data, according to the Code and Standards, it is:

A) permissible to leave details out in a brief presentation.

B) recommended that a minimum of five years performance history be included.

C) a requirement to present composite performance rather than individual account performance.

Explanation

Members and candidates should consider the knowledge and sophistication of those receiving the performance information. If only a brief presentation is given, members and candidates must make detailed information supporting the presentation available upon request. Individual account performance is permitted and no minimum number of years is recommended (these refer to GIPS, with which the Code and Standards do not require compliance). The primary requirement of Standard III(D) Performance Presentation is to not present false or misleading information. (Module 3.5)

For Further Reference:

(Study Session 1, Module 3.5, LOS 2.b, 2.c, LOS 3.a, 3.b, 3.c)

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Question #18 of 120

Question ID: 1207638

Bob Sampson is the head portfolio manager for Global Equities, which has been in existence for eight years. Beginning this year, the firm has decided to present performance information in compliance with GIPS. To claim GIPS compliance, the firm must present at least:

A) eight years of GIPS-compliant performance information.

B) **five years of GIPS-compliant performance information with no additional disclosure required for prior years.**

C) five years of GIPS-compliant performance information and may include noncompliant performance information for the prior three years in the “Disclosures” section.

Explanation

GIPS require that, to claim compliance, firms must present GIPS-compliant performance information for a minimum of five years or since inception if in existence less than five years. Firms may not link noncompliant performance information for any periods after January 1, 2000.

For Further Reference:

(Study Session 1, Module 4.1, LOS 4.a)

CFA® Program Curriculum, Volume 1, page 227

Question #19 of 120

Question ID: 1207641

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

An investor wants to receive $10,000 annually for ten years with the first payment five years from today. If the investor can earn a 14% annual return, the amount that she will have to invest today is *closest* to:

A) $27,091.

B) **$30,884.**

C) $52,161.

Explanation

This problem involves determining the present value of an annuity followed by finding the present value of a lump sum. Enter PMT = 10,000, N = 10, and I = 14. Compute PV = 52,161.16. That is the present value of the 10-year annuity, four years from today. Next, we need to discount that back to present for four years to find the amount of the investment today. Enter FV = −52,161.16, N = 4, I = 14, PMT = 0. Compute PV = 30,883.59.

For Further Reference:

(Study Session 2, Module 6.3, LOS 6.e)

CFA® Program Curriculum, Volume 1, page 316

Question #20 of 120

Question ID: 1207642

Which of the following statements about the frequency distribution shown below is *least accurate*?

Return Interval	Frequency
0% to 5%	10
> 5% to 10%	20
> 10% to 15%	30
> 15% to 20%	20

A) The return intervals are mutually exclusive.

B) **The cumulative absolute frequency of the fourth interval is 20.**

C) The relative frequency of the second return interval is 25%.

Explanation

The cumulative absolute frequency of the fourth interval is 80, which is the sum of the absolute frequencies from the first to the fourth intervals.

For Further Reference:

(Study Session 2, Module 7.1, LOS 7.b)

CFA® Program Curriculum, Volume 1, page 371

Question #21 of 120

Question ID: 1207644

An analyst obtains the following annual returns for a group of stocks: 10%, 8%, 7%, 9%, 10%, 12%, 11%, 10%, 30%, and 13%. This distribution:

A) has a median greater than its mode.

B) is skewed to the right, and the mean is less than the median.

C) **is skewed to the right, and the mean is greater than the mode.**

Explanation

Mean = 120 / 10 = 12, median = 10, and mode = 10. The distribution is skewed to the right, so the mean is greater than the median and mode. Generally, for positively skewed distributions, the mode is less than the median, which is less than the mean. In this case, the mode and median are equal because the number of observations is small.

For Further Reference:

(Study Session 2, Modules 7.2, 7.3, LOS 7.e, 7.k)

CFA® Program Curriculum, Volume 1, page 386

CFA® Program Curriculum, Volume 1, page 424

Question #22 of 120

Question ID: 1207645

An analyst gathers the following data about the mean monthly returns of three securities:

Security	Mean Monthly Return	Standard Deviation
X	0.9	0.7
Y	1.2	4.7
Z	1.5	5.2

Which security has the highest level of relative risk as measured by the coefficient of variation?

A) X.
B) **Y.**
C) Z.

Explanation

The coefficient of variation, CV = standard deviation / arithmetic mean, is a common measure of relative dispersion (risk). CV_X = 0.7 / 0.9 = 0.78; CV_Y = 4.7 / 1.2 = 3.92; and CV_Z = 5.2 / 1.5 = 3.47. Because a higher CV means higher relative risk, Security Y has the highest relative risk.

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.i)

CFA® Program Curriculum, Volume 1, page 422

Question #23 of 120

Question ID: 1207643

The median of a distribution is *least likely* equal to:

A) the second quartile.
B) **the third quintile.**
C) the fifth decile.

Explanation

The median is the midpoint of a distribution, such that 50% of the observations are greater than the median and 50% are less than the median. This is equivalent to the second quartile (4 groups) and the fifth decile (10 groups). If a distribution is divided into quintiles (5 groups), 60% of the observations are less than the third quintile.

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.f)

CFA® Program Curriculum, Volume 1, page 403

Question #24 of 120

Question ID: 1207646

Which of the following statements about probability concepts is *most accurate*?

A) **Subjective probability is a probability that is based on personal judgment.**
B) A conditional probability is the probability that two or more events happen concurrently.
C) An empirical probability is one based on logical analysis rather than on observation or personal judgment.

Explanation

Subjective probability is based on personal judgment. A *joint probability* is a probability that two or more events happen concurrently. An *a priori probability* is one based on logical analysis rather than on observation or personal judgment. An *empirical probability* is calculated using historical data. A *conditional probability* is the probability of one event happening on the condition that another event is certain to occur.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.a, 8.b, 8.d)

CFA® Program Curriculum, Volume 1, page 458

CFA® Program Curriculum, Volume 1, page 459

CFA® Program Curriculum, Volume 1, page 462

Question #25 of 120

Question ID: 1207652

For a test with sample size *n* of whether two variables are correlated, the critical values are based on:

A) *n* degrees of freedom.
B) *n* – 1 degrees of freedom.
C) ***n* – 2 degrees of freedom.**

Explanation

The test statistic for the hypothesis that correlation = 0 follows a *t*-distribution with *n* – 2 degrees of freedom.

For Further Reference:

(Study Session 3, Module 11.3, LOS 11.k)

CFA® Program Curriculum, Volume 1, page 589

Question #26 of 120

Question ID: 1207647

Alex White, CFA, is examining a portfolio that contains 100 stocks that are either value or growth stocks. Of these 100 stocks, 40% are value stocks. The previous portfolio manager had selected 70% of the value stocks and 80% of the growth stocks. What is the probability of selecting a stock at random that is either a value stock or was selected by the previous portfolio manager?

A) 28%.

B) 76%.

C) **88%.**

Explanation

	Selected by the Previous Portfolio Manager	Selected by the Current Portfolio Manager	Total
Value stocks	28 (28%)	12 (12%)	40
Growth stocks	48 (48%)	12 (12%)	60
Total	76	24	100

This problem involves the addition rule for probabilities, P(A or B) = P(A) + P(B) − P(AB).

P(A) is the probability that a randomly selected stock is a value stock, which is given as 40%.

P(B) is the probability that a stock was picked by the previous manager. That probability is (0.7)(0.4) + (0.8)(1 − 0.4) = 0.76. The previous manager selected 76% of the stocks in the portfolio.

P(AB) is the probability that a randomly selected stock is a value stock picked by the previous manager. Since the previous manager picked 70% of the 40% of the stocks that are value stocks, the probability that a randomly selected stock is a value stock picked by the previous manager is 40% × 70% = 28%.

From this, we have P(A or B) = 40% + 76% − 28% = 88%.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.f)

CFA® Program Curriculum, Volume 1, page 463

Question #27 of 120

Question ID: 1207648

Which of the following statements about the normal distribution is *least accurate*? The normal distribution:

A) **has a mean of zero and a standard deviation of one.**

B) is completely described by its mean and standard deviation.

C) is bell-shaped, with tails extending without limit to the left and to the right.

Explanation

The *standard* normal distribution has a mean of 0 and a standard deviation of 1.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.g and Study Session 3, Module 9.2, LOS 9.i)

CFA® Program Curriculum, Volume 1, page 467

CFA® Program Curriculum, Volume 1, page 531

Question #28 of 120

Question ID: 1207649

A manager forecasts a bond portfolio return of 10% and estimates a standard deviation of annual returns of 4%. Assuming a normal returns distribution and that the manager is correct, there is:

A) a 90% probability that the portfolio return will be between 3.2% and 17.2%.

B) a 95% probability that the portfolio return will be between 2.16% and 17.84%.

C) a 32% probability that the portfolio return will be between 6% and 14%.

Explanation

The 95% confidence interval is 10% ± 1.96(4%) or from 2.16% to 17.84%. The 90% confidence interval is 10% ± 1.65(4%) or from 3.4% to 16.6%. The probability of a return ± 1 standard deviation from the mean, between 6% and 14%, is approximately 68%.

For Further Reference:

(Study Session 3, Module 9.2, LOS 9.k)

CFA® Program Curriculum, Volume 1, page 533

Question #29 of 120

Question ID: 1207650

An investment has an expected return of 10% with a standard deviation of 5%. If the returns are normally distributed, the probability of losing money is *closest* to:

A) 2.5%.

B) 5.0%.

C) 16.0%.

Explanation

Using the standard normal probability distribution, $\mathrm{z} = \frac{\text{observation} - \text{mean}}{\text{standard deviation}} = \frac{0 - 10}{5} = -2.0$, the chance of getting zero or less return (losing money) is 1 − 0.9772 = 0.0228% or 2.28%. An alternative explanation: the expected return is 10%. To lose money means the return must fall below zero. Zero is about two standard deviations to the left of the mean. 50% of the time, a return will be below the mean, and 2.5% of the observations are below two standard deviations down. About 97.5% of the time, the return will be above zero. Thus, only about a 2.5% chance exists of having a value below zero.

For Further Reference:

(Study Session 3, Module 9.2, LOS 9.l)

CFA® Program Curriculum, Volume 1, page 534

Question #30 of 120

Question ID: 1207651

Which of the following statements about sampling and estimation is *least accurate*?

A) Sampling error is the difference between the observed value of a statistic and the value it is intended to estimate.

B) A simple random sample is a sample obtained in such a way that each element of the population has an equal probability of being selected.

C) **The central limit theorem states that the sample mean for a large sample size will have a distribution that is the same as the distribution of the underlying population.**

Explanation

According to the central limit theorem, the sample mean for large sample sizes will be distributed normally regardless of the distribution of the underlying population.

For Further Reference:

(Study Session 3, Module 10.1, LOS 10.c, 10.d, 10.e)

CFA® Program Curriculum, Volume 1, page 572

CFA® Program Curriculum, Volume 1, page 573

CFA® Program Curriculum, Volume 1, page 576

Question #31 of 120

Question ID: 1207659

Questions 31 through 42 relate to Economics. (18 minutes)

The crowding-out effect suggests that:

A) government borrowing will lead to an increase in private savings.

B) as government spending increases, so will incomes and taxes, and the higher taxes will reduce both aggregate demand and output.

C) **greater government deficits will drive up interest rates, thereby reducing private investment.**

Explanation

The crowding-out effect refers to a reduction in private borrowing and investment as a result of higher interest rates generated by budget deficits that are financed by borrowing in the private loanable funds market.

For Further Reference:

(Study Session 5, Module 16.3, LOS 16.q)

CFA® Program Curriculum, Volume 2, page 305

Question #32 of 120

Question ID: 1207664

Consider two currencies, the WSC and the BDR. The spot WSC/BDR exchange rate is 2.875, the 180-day riskless WSC rate is 1.5%, and the 180-day riskless BDR rate is 3.0%. The 180-day forward exchange rate that will prevent arbitrage profits is *closest* to:

A) 2.833 WSC/BDR.

B) **2.854 WSC/BDR.**

C) 2.918 WSC/BDR.

Explanation

Arbitrage-free forward = 2.875 WSC/BDR × [(1 + 0.015 / 2) / (1 + 0.03 / 2)] = 2.8538 WSC/BDR.

For Further Reference:

(Study Session 5, Module 18.2, LOS 18.h)

CFA® Program Curriculum, Volume 2, page 423

Question #33 of 120

Question ID: 1207654

An advantage of the Herfindahl-Hirschman Index (HHI) over the N-firm concentration ratio as a summary measure of the market structure of an industry is that the HHI is more sensitive to:

A) **mergers.**
B) barriers to entry.
C) elasticity of demand.

Explanation

The HHI is more sensitive to the effects of mergers compared to the N-firm concentration ratio. Neither measure accounts for elasticity of demand or barriers to entry.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.g)

CFA® Program Curriculum, Volume 2, page 103

Question #34 of 120

Question ID: 1207658

Which of the following does the U.S. central bank *most often* use to change the money supply?

A) The discount rate.
B) **Open market operations.**
C) The required reserve ratio.

Explanation

Open market operations are the U.S. Federal Reserve's most often used tool for changing the money supply.

For Further Reference:

(Study Session 5, Module 16.2, LOS 16.h)

CFA® Program Curriculum, Volume 2, page 281

Question #35 of 120

Question ID: 1207653

The price of milk in a country increases from €1.00 per liter to €1.10 per liter, and the quantity supplied does not change. This suggests the elasticity of the short-run supply of milk in this country is equal to:

A) infinity, and supply is perfectly elastic.

B) **zero, and supply is perfectly inelastic.**

C) infinity, and supply is perfectly inelastic.

Explanation

If quantity supplied does not respond to a change in price, supply is perfectly inelastic. For perfectly inelastic supply, elasticity equals zero.

For Further Reference:

(Study Session 4, Module 12.1, LOS 12.a)

CFA® Program Curriculum, Volume 2, page 9

Question #36 of 120

Question ID: 1207657

In the short run, will an increase in the money supply increase the price level and real output?

A) **Both will increase in the short run.**

B) Neither will increase in the short run.

C) Only one will increase in the short run.

Explanation

In the short run, an increase in the money supply will increase aggregate demand. The new short-run equilibrium will be at a higher price level and a greater level of real output (GDP).

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.i)

CFA® Program Curriculum, Volume 2, page 161

Question #37 of 120

Question ID: 1207662

The country of Colfax can produce 15 units of rice or 10 units of plastic per day of labor. The country of Birklund can produce 18 units of rice or 12 units of plastic per day of labor. With regard to potential benefits of trading rice and plastic between Colfax and Birklund:

A) **there are no potential gains from trade.**

B) Colfax should produce and trade rice for Birklund's plastic.

C) Birklund should produce and trade rice for Colfax's plastic.

Explanation

In this case, there are no clear potential benefits from trade because the countries' opportunity costs of production are equal. Colfax's opportunity cost of rice = 10 / 15 = 0.67 units of plastic, and Birklund's opportunity cost of rice = 12 / 18 = 0.67 units of plastic. Colfax's opportunity cost of plastic = 15 / 10 = 1.5 units of rice, and Birklund's opportunity cost of plastic = 18 / 12 = 1.5 units of rice.

For Further Reference:

(Study Session 5, Module 17.1, LOS 17.c)

CFA® Program Curriculum, Volume 2, page 343

Question #38 of 120

Question ID: 1207661

The public sector is *most likely* to increase as a proportion of economic output if fiscal policy:

A) and monetary policy are both expansionary.

B) is contractionary and monetary policy is expansionary.

C) **is expansionary and monetary policy is contractionary.**

Explanation

Reducing taxes is likely to keep the public sector size unchanged, but an increase in government spending will increase the size of the public sector. Contractionary monetary policy is likely to decrease the output of the private sector as interest rates rise.

For Further Reference:

(Study Session 5, Module 16.3, LOS 16.t)

SchweserNotes, Book 2 page 125

CFA® Program Curriculum, Volume 2, page 319

Question #39 of 120

Question ID: 1207663

Unlike members of free trade areas, customs union members:

A) adopt a single currency.

B) remove barriers to trade with all members.

C) **adopt uniform trade restrictions with non-members.**

Explanation

Customs unions adopt uniform trade restrictions with non-members. Customs union members do not adopt a single currency. Both free trade areas and customs unions remove trade barriers among their members.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.f)

SchweserNotes, Book 2 page 144

CFA® Program Curriculum, Volume 2, page 358

Question #40 of 120

Question ID: 1207656

A perfectly elastic aggregate supply curve represents:

A) the productive capacity of an economy at full employment.

B) the production decisions of firms only in the very short run.

C) the short-run relationship between output and the price level.

Explanation

The very short run aggregate supply curve is perfectly elastic because firms can adjust output by increasing or decreasing labor hours and capacity use without affecting input prices. The short-run aggregate supply curve is upward sloping. The long-run aggregate supply curve is perfectly inelastic and represents potential GDP, the full-employment output level of an economy.

For Further Reference:

(Study Session 4, Module 14.2, LOS 14.g)

SchweserNotes, Book 2 page 61

CFA® Program Curriculum, Volume 2, page 147

Question #41 of 120

Question ID: 1207660

Reasons why the unemployment rate is a lagging indicator of the business cycle *least likely* include:

A) discouraged workers who begin seeking work.

B) action lag in the implementation of unemployment insurance.

C) high costs to employers of frequently hiring or firing employees.

Explanation

Unemployment insurance is an example of an automatic stabilizer that is not subject to the action lag of discretionary fiscal policy tools. One reason why the unemployment rate is a lagging indicator is the fact that employers are slow to lay off employees early in recessions and slow to add employees early in expansions, because frequent hiring and firing has high costs. Another reason is that early in expansions, more discouraged workers (who are not counted as unemployed because they are out of the labor force) may begin seeking work (thereby re-entering the labor force) than the number of new jobs that are available, which increases the unemployment rate.

For Further Reference:

(Study Session 4, Modules 15.1, 15.2, LOS 15.b, 15.d, 15.i, Study Session 5, Module 16.3, LOS 16.p)

SchweserNotes, Book 2 page 84

CFA® Program Curriculum, Volume 2, page 200

SchweserNotes, Book 2 page 88

CFA® Program Curriculum, Volume 2, page 219

SchweserNotes, Book 2 page 96

CFA® Program Curriculum, Volume 2, page 237

SchweserNotes, Book 2 page 120

CFA® Program Curriculum, Volume 2, page 309

Question #42 of 120

Question ID: 1207655

A natural monopoly is *most likely* to exist when:

A) economies of scale are great.

B) average total cost increases as output increases.

C) a single firm owns essentially all of a productive resource.

Explanation

A natural monopoly may exist when economies of scale are great. The large economies of scale mean that a single producer results in the lowest production costs.

For Further Reference:

(Study Session 4, Modules 13.1, 13.4, LOS 13.a, 13.b)

SchweserNotes, Book 2 page 20

CFA® Program Curriculum, Volume 2, page 64

SchweserNotes, Book 2 page 22

CFA® Program Curriculum, Volume 2, page 64

Question #43 of 120

Question ID: 1207668

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

A company has the following sequence of events regarding its stock:

- The company had 1,000,000 shares outstanding at the beginning of the year.
- On June 30, the company declared and issued a 10% stock dividend.
- On September 30, the company sold 400,000 shares of common stock at par.

The number of shares that should be used to compute basic earnings per share at year end is:

A) 1,000,000.

B) 1,100,000.

C) 1,200,000.

Explanation

original shares of common stock	= 1,000,000(12)	= 12,000,000
stock dividend	= 100,000(12)	= 1,200,000
new shares of common stock	= 400,000(3)	= 1,200,000
total shares of common stock	$=\dfrac{14,400,000}{12}$	= 1,200,000

Stock dividends are assumed to have been outstanding since the beginning of the year.

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g)

SchweserNotes, Book 3 page 56

CFA® Program Curriculum, Volume 3, page 99

Question #44 of 120

Question ID: 1207671

Time-series analysis of a firm's common-size balance sheets reveals the following data:

	20X3	20X4	20X5
Current assets	20%	22%	25%
Inventory	8%	9%	11%
Short-term debt	10%	11%	12%
Long-term debt	24%	21%	18%

Based only on the data provided, an analyst can conclude that the firm's:

A) debt ratio is decreasing.

B) quick ratio is decreasing.

C) inventory/sales ratio is increasing.

Explanation

The debt ratio is total debt to total assets. Because common-size balance sheet data are stated as percentages of total assets, the debt ratio can be determined from the data given.

20X3: 10% + 24% = 34%

20X4: 11% + 21% = 32%

20X5: 12% + 18% = 30%

The debt ratio is decreasing over the period shown. Neither the inventory/sales ratio nor the quick ratio can be determined from the data given because the data do not include sales or current liabilities.

For Further Reference:

(Study Session 7, Module 22.7, LOS 22.g, 22.h)

SchweserNotes, Book 3 page 92

CFA® Program Curriculum, Volume 3, page 161

SchweserNotes, Book 3 page 94

CFA® Program Curriculum, Volume 3, page 169

Question #45 of 120

Question ID: 1207672

Which of the following statements about the analysis of cash flows is *least accurate*?

A) Interest payments on debt are not a financing cash flow under U.S. GAAP.

B) **Both the direct and indirect methods involve adding back noncash items such as depreciation and amortization.**

C) When using the indirect method, an analyst should add any losses on the sales of fixed assets to net income.

Explanation

When using the direct method of calculating operating cash flows, depreciation and amortization are not "added back" (to net income) because we don't begin with net income under the direct method. Depreciation and amortization are noncash changes and are not used under the direct method. The other statements are true. Interest payments on debt affect cash flow from operations. When using the indirect method, an analyst should add any losses on sales of fixed assets to net income since they are not operating cash flows.

For Further Reference:

(Study Session 7, Modules 23.1, 23.2, LOS 23.c, 23.f)

SchweserNotes, Book 3 page 105

CFA® Program Curriculum, Volume 3, page 187

SchweserNotes, Book 3 page 109

CFA® Program Curriculum, Volume 3, page 199

Question #46 of 120

Question ID: 1207676

A company that reports under U.S. GAAP and changes its inventory cost assumption from weighted average cost to last-in first-out is required to apply this change in accounting principle:

A) retrospectively, and disclose the new cost flow method being used.

B) **prospectively, and explain the reasons for the change in the financial statement disclosures.**

C) retrospectively, and explain the reasons for the change in the financial statement disclosures.

Explanation

Under U.S. GAAP, a change to LIFO from another inventory cost method is an exception to the requirement of retrospective application of changes in an accounting principle. Instead of restating prior years' data, the firm uses the carrying value of inventory at the time of the change as the first LIFO layer. U.S. GAAP requires a company that is changing its inventory cost assumption to explain, in its financial statement disclosures, why the new method is preferable to the old method.

For Further Reference:

(Study Session 8, Module 25.4, LOS 25.i)

SchweserNotes, Book 3 page 193

CFA® Program Curriculum, Volume 3, page 339

Question #47 of 120

Question ID: 1207721

For which of the following investments in securities is a firm *most likely* to report unrealized gains or losses on its income statement?

A) Preferred stock, which the firm classifies as available-for-sale.

B) Five-year bonds, which the firm purchased in a private placement.

C) Listed call options, which the firm intends to exercise at expiration.

Explanation

Options are derivatives, which are reported at fair value on the balance sheet with unrealized gains and losses recognized on the income statement. Available-for-sale securities are marked to market on the balance sheet, but unrealized gains and losses are reported in owners' equity as other comprehensive income. Bonds purchased in a private placement cannot be resold to the public and therefore are likely to be classified as held-to-maturity, in which case the firm does not recognize unrealized gains or losses.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e, Study Session 12, Module 36.3, LOS 36.i and Study Session 16, Module 48.1, LOS 48.a)

SchweserNotes, Book 3 page 82

CFA® Program Curriculum, Volume 3, page 133

SchweserNotes, Book 4 page 205

CFA® Program Curriculum, Volume 4, page 252

SchweserNotes, Book 5 page 158

CFA® Program Curriculum, Volume 5, page 386

Question #48 of 120

Question ID: 1207667

Consider a manufacturing company and a financial services company. Interest expense is *most likely* classified as a non-operating component of net income for:

A) both of these companies.

B) neither of these companies.

C) only one of these companies.

Explanation

Interest expense is shown as a non-operating component of net income for a manufacturing company but would typically be classified as an operating expense for a financial services company.

For Further Reference:

(Study Session 7, Module 21.3, LOS 21.f)

SchweserNotes, Book 3 page 56

CFA® Program Curriculum, Volume 3, page 98

Question #49 of 120

Question ID: 1207666

The two primary assumptions in preparing financial statements under IFRS are:

A) accrual accounting and going concern.

B) reasonable accuracy and accrual accounting.

C) going concern and reasonable accuracy.

Explanation

In the IFRS framework, the two assumptions that underlie the preparation of financial statements are accural accounting and the going concern assumption.

For Further Reference:

(Study Session 6, Module 20.2, LOS 20.c)

SchweserNotes, Book 3 page 28

CFA® Program Curriculum, Volume 3, page 54

Question #50 of 120

Question ID: 1207681

Forman, Inc., and Swoft, Inc., both operate within the same industry. Forman's stated strategy is to differentiate its premium products relative to its competitors, while Swoft is a low-cost producer. Given the companies' stated strategies, Forman *most likely* has:

A) higher gross margins relative to Swoft.
B) lower advertising expenses relative to Swoft.
C) lower research and development expenses relative to Swoft.

Explanation

An analyst can use the historical trend in a firm's financial ratios as well as an industry relative comparison to assess the firm's business strategy. A firm producing premium products with a strategy of differentiation should have higher gross margins, higher advertising expenses, and higher research and development expenses relative to firms in its industry that pursue a low-cost-of-production strategy.

For Further Reference:

(Study Session 9, Module 30.1, LOS 30.a)

SchweserNotes, Book 3 page 322

CFA® Program Curriculum, Volume 3, page 609

Question #51 of 120

Question ID: 1207669

An analyst gathered the following data about a company:

- 1,000 common shares are outstanding (no change during the year).
- Net income is $5,000.
- The company paid $500 in preferred dividends.
- The company paid $600 in common dividends.
- The average market price of their common stock is $60 for the year.
- The company had 100 warrants (for one share each) outstanding for the entire year, exercisable at $50.

The company's diluted earnings per share is *closest* to:

A) $4.42.
B) $4.55.

C) $4.83.

Explanation

The warrants are dilutive because their exercise price is less than the average market price.

shares issued to warrant holders = 100

warrants generate cash of 100(50) = $5,000

$$\text{repurchased shares} = \frac{5{,}000}{60} = 83$$

net new shares created = 100 − 83 = 17

Alternatively, $\frac{60-50}{60} \times 100 \approx 17$

$$\text{diluted EPS} = \frac{\text{NI} - \text{preferred dividends}}{\text{weighted average \# shares} + \text{warrant adjust}}$$

$$\text{diluted EPS} = \frac{5{,}000-500}{1{,}017} = \$4.42$$

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g, 21.h)

SchweserNotes, Book 3 page 56

CFA® Program Curriculum, Volume 3, page 99

SchweserNotes, Book 3 page 56

CFA® Program Curriculum, Volume 3, page 131

Question #52 of 120

Question ID: 1207665

Which of the following sources of information should an analyst consider the *least* reliable?

A) Form 10-Q.

B) Proxy statement.

C) **Corporate press release.**

Explanation

Corporate press releases are written by management and are often viewed as public relations or sales materials because of the great possibility of inherent management bias in such documents. Often, little or none of the material is independently reviewed by outside auditors. Such documents are not mandated by the securities regulators. Form 10-Q (quarterly financial statements) and proxy statements are mandatory SEC filings in the United States, which inherently increases their reliability given the penalties that can be imposed by the SEC if any serious irregularities are subsequently found.

For Further Reference:

(Study Session 6, Module 19.2, LOS 19.e)

SchweserNotes, Book 3 page 4

CFA® Program Curriculum, Volume 3, page 31

Question #53 of 120

Question ID: 1207678

A company takes a $10 million impairment charge on a depreciable asset in 20X3. The *most likely* effect will be to:

A) increase reported net income in 20X4.

B) decrease net income and taxes payable in 20X3.

C) increase return on equity and operating cash flow in 20X4.

Explanation

The impairment writedown in 20X3 will reduce depreciation expense in 20X4, which will increase 20X4 EBIT and net income. Operating cash flow and taxes payable are not affected because an impairment cannot be deducted from income for tax reporting purposes until the asset is sold or otherwise disposed of.

For Further Reference:

(Study Session 8, Module 26.3, LOS 26.i)

SchweserNotes, Book 3 page 223

CFA® Program Curriculum, Volume 3, page 411

Question #54 of 120

Question ID: 1207679

Xanos Corporation faced a 50% marginal tax rate last year and showed the following financial and tax reporting information:

- Deferred tax asset of $1,000.
- Deferred tax liability of $5,000.

Based only on this information and the news that the tax rate will decline to 40%, Xanos Corporation's deferred tax:

A) asset will be reduced by $400 and deferred tax liability will be reduced by $2,000.

B) liability will be reduced by $1,000 and income tax expense will be reduced by $800.

C) asset will be reduced by $200 and income tax expense will be reduced by $1,000.

Explanation

There is a 20% reduction in the tax rate [(40% − 50%) / 50% = –0.2]. Hence, the deferred tax asset will be $800 = $1,000(1 − 0.2), the deferred tax liability will be $4,000 = $5,000(1 − 0.2), and the income tax expense will fall by the net amount of the decline in the asset and liability balances ($1,000 – $200 = $800).

For Further Reference:

(Study Session 8, Module 27.4, LOS 27.d)

SchweserNotes, Book 3 page 249

CFA® Program Curriculum, Volume 3, page 454

Question #55 of 120

Question ID: 1207680

The ratio of operating cash flow to net income is *most likely* to indicate low quality of earnings when it is:

A) **less than one.**

B) highly variable.

C) increasing over time.

Explanation

Operating cash flow that is less than net income (ratio less than one) or declining over time may indicate low-quality earnings from aggressive accounting or accounting irregularities. A ratio of operating cash flow to net income that is highly variable, but consistently greater than one, is not necessarily indicative of low-quality earnings.

For Further Reference:

(Study Session 9, Module 29.3, LOS 29.i)

SchweserNotes, Book 3 page 315

CFA® Program Curriculum, Volume 3, page 591

Question #56 of 120

Question ID: 1207677

A company that capitalizes costs instead of expensing them will have:

A) higher income variability and higher cash flows from operations.

B) **lower cash flows from investing and lower income variability.**

C) lower cash flows from operations and higher profitability in early years.

Explanation

Capitalizing costs tends to smooth earnings and reduces investment cash flows. It will also increase cash flows from operations and increase profitability in the early years.

For Further Reference:

(Study Session 8, Module 26.1, LOS 26.c)

SchweserNotes, Book 3 page 213

CFA® Program Curriculum, Volume 3, page 386

Question #57 of 120

Question ID: 1207674

Al Pike, CFA, is analyzing Red Company by projecting pro forma financial statements. Pike expects Red to generate sales of $3 billion and a return on equity of 15% in the next year. Pike forecasts that Red's total assets will be $5 billion and that the company will maintain its financial leverage ratio of 2.5. Based on these forecasts, Pike should project Red's net income to be:

A) $100 million.

B) **$300 million.**

C) $500 million.

Explanation

Based on the data given, use the basic DuPont equation and solve for expected net income.

ROE = (net income / revenues) × (revenues / total assets) × (total assets / total equity)

0.15 = (net income / $3 billion) × ($3 billion / $5 billion) × 2.5

net income / $3 billion = 0.1

net income = $300 million

Alternatively, A/E = 2.5 and assets = $5 billion, so equity = $2 billion.

net income / equity = 15%, so net income = 0.15($2 billion) = $300 million.

For Further Reference:

(Study Session 7, Module 24.4, LOS 24.d)

SchweserNotes, Book 3 page 323

CFA® Program Curriculum, Volume 3, page 612

Question #58 of 120

Question ID: 1207670

Which of the following items would affect owners' equity and also appear on the income statement?

A) Dividends paid to shareholders.
B) **Unrealized gains and losses on trading securities.**
C) Unrealized gains and losses on available-for-sale securities.

Explanation

Unrealized gains and losses from trading securities are reflected in the income statement and affect owners' equity. However, unrealized gains and losses from available-for-sale securities are included in other comprehensive income. Transactions included in other comprehensive income affect equity but not net income. Dividends paid to shareholders reduce owners' equity but not net income.

For Further Reference:

(Study Session 7, Module 21.5, LOS 21.l)

SchweserNotes, Book 3 page 67

CFA® Program Curriculum, Volume 3, page 145

Question #59 of 120

Question ID: 1207673

Copper, Inc., had $4 million in bonds outstanding that were convertible into common stock at a conversion rate of 100 shares per $1,000 bond. In 20X1, all of the outstanding bonds were converted into common stock. Copper's average share price for 20X1 was $15. Copper's statement of cash flows for the year ended December 31, 20X1, should *most likely* include:

A) **a footnote describing the conversion of the bonds into common stock.**
B) cash flows from financing of +$4 million from issuance of common stock and –$4 million from retirement of bonds.
C) cash flows from financing of +$6 million from issuance of common stock and –$4 million from retirement of bonds and cash flows from investing of –$2 million for a loss on retirement of bonds.

Explanation

Conversion of bonds into common stock is a non-cash transaction, but the conversion should be disclosed in a footnote to the statement of cash flows.

For Further Reference:

(Study Session 7, Module 23.2, LOS 23.f)

SchweserNotes, Book 3 page 109

CFA® Program Curriculum, Volume 3, page 199

Question #60 of 120

Question ID: 1207675

During a period of falling costs of manufacturing, which of the following inventory cost formulas would result in the greatest reported net income?

A) **LIFO.**
B) FIFO.
C) Average cost.

Explanation

With LIFO, more recent, lower costs would be used for COGS. A reduction in COGS will increase gross profits and net income, other things equal.

For Further Reference:

(Study Session 8, Module 25.1, LOS 25.b)

SchweserNotes, Book 3 page 179

CFA® Program Curriculum, Volume 3, page 314

Question #61 of 120

Question ID: 1207685

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

An analyst identifies the following cash flows for an average-risk project:

Year 0	−$5,000
Year 1-2	$1,900
Year 3	$2,500
Year 4	$2,000

If the company's cost of capital is 12%, the project's discounted payback period is *closest* to:

A) 2.5 years.
B) **3.0 years.**
C) 3.9 years.

Explanation

Year	0	1	2	3	4
Net cash flow	−$5,000.00	$1,900.00	$1,900.00	$2,500.00	$2,000.00
Net cash flow discounted at 12%	−$5,000.00	$1,696.43	$1,514.67	$1,779.45	$1,271.04
Cumulative discounted net cash flows	−$5,000.00	−$3,303.57	−$1,788.90	−$9.45	$1,261.59

$$\text{discounted payback period} = 3 + \frac{9.45}{1{,}271.04} = 3.01 \text{ years}$$

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.d)

SchweserNotes, Book 4 page 21

CFA® Program Curriculum, Volume 4, page 52

Question #62 of 120

Question ID: 1207691

Mary Miller, CFA, manages the short-term cash position for Young Company. Miller can invest in one of three securities that will mature in 180 days: a Treasury bill priced at 97.5% of par, commercial paper with a bond-equivalent yield of 5.10%, and a 6-month certificate of deposit that will return 2.5% over the 180-day holding period. Miller should purchase:

A) the Treasury bill.
B) the commercial paper.
C) the certificate of deposit.

Explanation

To compare these short-term securities, state the return on each as a bond-equivalent yield. For the commercial paper, the BEY is given as 5.10%. The BEY for the certificate of deposit is 2.5% × 365 / 180 = 5.07%. For the Treasury bill, the BEY is [(100 − 97.5) / 97.5] × 365 / 180 = 5.20%. Miller should purchase the Treasury bill.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.e)

SchweserNotes, Book 4 page 85

CFA® Program Curriculum, Volume 4, page 169

Question #63 of 120

Question ID: 1207687

Which of the following is the *least appropriate* method for estimating a firm's before-tax cost of debt capital?

A) Use the market yield on bonds with a rating and maturity similar to the firm’s existing debt.
B) Assume the firm’s cost of debt capital is equal to the yield to maturity on its publicly traded debt.
C) Use the coupon rate on the firm’s most recently issued debt.

Explanation

Current market yields, not the coupon rate, should be used to estimate the cost of debt capital.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.f)

SchweserNotes, Book 4 page 45

CFA® Program Curriculum, Volume 4, page 84

Question #64 of 120

Question ID: 1207689

Paul Dufray, CFA, is estimating the asset beta for a new project based on a firm that primarily makes and sells a similar product. In addition to the beta of that firm, Dufray will need to estimate the firm's:

A) sales risk and financial risk.
B) **debt-to-equity ratio and tax rate.**
C) operating leverage and financial leverage.

Explanation

To calculate the asset beta, the analyst will need to estimate the firm's beta, its debt-to-equity ratio, and its tax rate.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.i)

SchweserNotes, Book 4 page 49

CFA® Program Curriculum, Volume 4, page 94

Question #65 of 120

Question ID: 1207682

Stakeholder theory is *most accurately* described as the belief that corporate governance should focus on managing:

A) the activities of a company in the best interests of its owners.
B) employee activities in compliance with ethical standards and applicable laws.
C) **conflicts among different groups that have an interest in a company's activities.**

Explanation

According to stakeholder theory, the focus of corporate governance is managing conflicts among the interests of different groups that have an interest in a company's activities. Shareholder theory holds that the focus of corporate governance should be to manage the activities of a company in the best interests of its owners.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.a)

SchweserNotes, Book 4 page 1

CFA® Program Curriculum, Volume 4, page 6

Question #66 of 120

Question ID: 1207692

Which of the following working capital management outcomes is *least* desirable?

A) Low operating cycle.

B) High inventory turnover.

C) **High cash conversion cycle.**

Explanation

The cash conversion cycle measures the amount of time it takes for the firm to turn the firm's cash investments in inventory back into cash. A high cash conversion cycle implies that the company has too much invested in working capital.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.b)

SchweserNotes, Book 4 page 82

CFA® Program Curriculum, Volume 4, page 159

Question #67 of 120

Question ID: 1207686

Which of the following is *least likely* a problem associated with the internal rate of return (IRR) method of choosing investment projects?

A) Using IRR to rank mutually exclusive projects assumes reinvestment of cash flows at the IRR.

B) **For independent projects, the IRR and NPV can lead to different investment decisions.**

C) If the project has an unconventional cash flow pattern, the result can be multiple IRRs.

Explanation

IRR and NPV lead to the same decision when choosing independent projects but may lead to different decisions when choosing between projects.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.e)

SchweserNotes, Book 4 page 29

CFA® Program Curriculum, Volume 4, page 58

Question #68 of 120

Question ID: 1207690

Break points in a firm's marginal cost of capital schedule are *best* interpreted as representing:

A) the maximum amounts of debt, preferred stock, and common stock the firm can issue.

B) the amounts of new securities a firm would need to issue to take advantage of flotation cost discounts.

C) **the amounts of capital expenditure at which the company’s weighted average cost of capital increases.**

Explanation

Break points in the marginal cost of capital structure occur at the levels where the cost of one of the components of the company's capital structure increases, increasing its weighted average cost of capital.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.k)

SchweserNotes, Book 4 page 52

CFA® Program Curriculum, Volume 4, page 102

Question #69 of 120

Question ID: 1207683

With regard to environmental, social, and governance (ESG) considerations, which of the following statements is *most accurate*?

A) Fiduciary duty requires managers to integrate their clients' ESG-related considerations into investment decisions.

B) **Integrating ESG factors into the analysis of a company's risk and return characteristics is not considered a violation of fiduciary duty.**

C) A "values-based" objective involves investing in companies that have ESG-related opportunities that are not fully reflected in their share prices.

Explanation

Using ESG factors in estimating the risk and returns of a company is not considered a violation of a manager's fiduciary duty to clients and beneficiaries. ESG considerations may conflict with fiduciary duty if they result in the manager accepting lower returns or higher risk than they otherwise would. A "*values*-based" investment objective is to express the investor's ethical beliefs through investment decisions. A "*value*-based" approach to ESG investing refers to considering ESG-related risks and opportunities alongside traditional investment considerations.

For Further Reference:

(Study Session 10, Module 31.2, LOS 31.j)

SchweserNotes, Book 4 page 11

CFA® Program Curriculum, Volume 4, page 36

Question #70 of 120

Question ID: 1207688

Janet Adams, CFA, is reviewing Rival Company's financial statements. Rival's long-term debt totals $35 million, while total shareholder equity equals $140 million. Rival's long-term debt has a YTM of 9%. Rival's tax rate is 40% and its beta is 0.9. Adams gathers the following additional facts:

- Treasury bills earn 4.0%.
- The equity risk premium is 4.5%.

Based on the information provided, Rival's weighted average cost of capital is *closest* to:

A) 4.6%.

B) **7.5%.**

C) 8.2%.

Explanation

Weight of debt = 35 / (35 + 140) = 0.2

Weight of equity = 140 / (35 + 140) = 0.8

After-tax cost of debt = 9%(1 − 0.4) = 5.4%

Cost of equity = 4% + 0.9(4.5%) = 8.05%

WACC = 0.2(5.4%) + 0.8(8.05%) = 7.52%

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.a)

CFA® Program Curriculum, Volume 4, page 78

Question #71 of 120

Question ID: 1207684

Bear Company produces gravel-hauling equipment. The company recently began producing the Mauler, a new line of equipment. Prior to beginning production of the Mauler, the company spent $10 million in research and development costs. Bear expects the Mauler line to generate positive cash flows beginning in the fourth year. However, Bear is forecasting a one-time expense in year 5 to comply with new government emission standards. The company will use an empty building it already owns to produce the Mauler. When analyzing the project cash flows for the Mauler, Bear should *least appropriately* include:

A) the use of the empty building.

B) the research and development cost.

C) the compliance cost for emissions standards.

Explanation

The R&D expenditure is a sunk cost that should not be considered in the project's cash flows. The opportunity cost of the empty building in its next-best use should be considered in the project analysis.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.b)

CFA® Program Curriculum, Volume 4, page 50

Question #72 of 120

Question ID: 1207693

An analyst has calculated the following statistics for Company X and Company Y.

	Company X		Company Y	
	Year 1	Year 2	Year 1	Year 2
Number of days of inventory	18	22	33	24
Number of days of receivables	14	16	14	12
Number of days of payables	19	20	18	20

The net operating cycle for:

A) Company Y was 16 days in year 2, an improvement in liquidity compared to year 1.

B) Company Y was 36 days in year 2, a decline in liquidity compared to year 1.

C) Company X was 18 days in year 2, an improvement in liquidity compared to year 1.

Explanation

Net operating cycle is calculated as the number of days of inventory + number of days of receivables − number of days of payables. Company Y's net operating cycles were 33 + 14 − 18 = 29 days in year 1 and 24 + 12 − 20 = 16 days in year 2. The decline in net operating cycle days in year 2 indicates an improvement in liquidity. For Company X, the net operating cycle for year 2 was 22 + 16 − 20 = 18 days, an increase from year 1, which was 18 + 14 − 19 = 13 days.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.c)

CFA® Program Curriculum, Volume 4, page 159

Question #73 of 120

Question ID: 1207695

Questions 73 through 86 relate to Equity Investments. (21 minutes)

Which of the following statements about types of orders is *least accurate*?

A) Market orders are orders to buy or sell at the best price available.

B) Limit orders are orders to buy or sell at or away from the market price.

C) A stop buy order is typically used to protect a short position in a security and is placed below the current market price.

Explanation

A *stop buy order* is a conditional market order by which an investor directs the purchase of a stock if it rises to a certain price. Stop buys are placed above the current market price. Limit orders can have market price as the limit, used when lack of liquidity is a concern.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.g, 36.h)

SchweserNotes, Book 4 page 201

CFA® Program Curriculum, Volume 4, page 246

SchweserNotes, Book 4 page 201

CFA® Program Curriculum, Volume 4, page 246

Question #74 of 120

Question ID: 1207706

A stock has the following data associated with it:

- A required rate of return of 14%.
- A return on equity of 15%.
- An earnings retention rate of 40%.

The stock's justified price-to-earnings ratio is *closest* to:

A) 5.0.
B) 6.7.
C) **7.5.**

Explanation

$$P/E = \frac{\text{dividend payout ratio}}{k - g}$$

dividend payout ratio = 1 − retention ratio = 1 − 0.4 = 0.6

growth rate (g) = retention rate × ROE = 0.4 × 15% = 6%

$$P/E = \frac{0.6}{0.14 - 0.06} = 7.5$$

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.j)

SchweserNotes, Book 4 page 297

CFA® Program Curriculum, Volume 4, page 468

Question #75 of 120

Question ID: 1207704

The required rate of return used in the dividend discount model is *least likely* to be affected by a change in:

A) the expected rate of inflation.
B) the real risk-free rate of return.
C) **the growth rate of earnings and dividends.**

Explanation

The expected growth rate in dividends is an input into the dividend discount model, but the real risk-free rate, the expected inflation rate, and the risk premium are the components of the required rate of return.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.e)

SchweserNotes, Book 4 page 286

CFA® Program Curriculum, Volume 4, page 453

Question #76 of 120

Question ID: 1207702

High return on invested capital and high pricing power are *most likely* to be associated with an industry that has:

A) high capacity.
B) low barriers to entry.
C) **high concentration.**

Explanation

High return on invested capital and high pricing power are associated with high industry concentration (i.e., small number of firms), high barriers to entry, and low industry capacity.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.g)

SchweserNotes, Book 4 page 270

CFA® Program Curriculum, Volume 4, page 405

Question #77 of 120

Question ID: 1207699

With regard to the implications of stock market efficiency for technical analysis and fundamental analysis, if market prices are:

A) **weak-form efficient, technical analysis that depends only on past trading data should be of limited or no value.**
B) semistrong-form efficient, fundamental analysis using the top-down approach should yield consistently superior returns.
C) semistrong-form efficient, fundamental analysis using only publicly available market information should generate abnormal returns after considering risk and transaction costs.

Explanation

If capital markets are weak-form efficient and semistrong-form efficient, no publicly available information can be used to earn abnormal (risk-adjusted) returns.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

SchweserNotes, Book 4 page 240

CFA® Program Curriculum, Volume 4, page 330

Question #78 of 120

Question ID: 1207705

The following data pertains to a firm's common stock:

- The stock will pay no dividends for two years.
- The dividend three years from now is expected to be $1.
- Dividends are expected to grow at a 7% rate from that point onward.

If an investor requires a 17% return on this investment, how much will the investor be willing to pay for this stock now?

A) $6.24.
B) **$7.31.**
C) $8.26.

Explanation

Time line = $0 now; $0 in year 1; $0 in year 2; $1 in year 3.

$$P_2 = \frac{D_3}{(k-g)} = \frac{1}{(0.17-0.07)} = \$10$$

Note that the price is always one year before the dividend date. Solve for the PV of $10 to be received in two years.

FV = 10; N = 2; I = 17; CPT → PV = $7.31.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

SchweserNotes, Book 4 page 291

CFA® Program Curriculum, Volume 4, page 459

Question #79 of 120

Question ID: 1207707

A stock's price currently is $100. An analyst forecasts the following for the stock:

- The normalized trailing price earnings (P/E) ratio will be 12×.
- The stock is expected to pay a $5 dividend this coming year on projected earnings of $10 per share.

If the analyst were to buy and hold the stock for the year, the projected rate of return based on these forecasts is *closest* to:

A) 15%.
B) 20%.
C) **25%.**

Explanation

The forecast year-end price, *P*, is:

$$P = EPS \times (P/E) = 10\,(12) = 120$$

$$\begin{aligned} \text{expected return} &= \frac{\text{dividend}+(\text{ending price}-\text{beginning price})}{\text{beginning price}} \\ &= \frac{\$5+\$120-\$100}{\$100} \\ &= 0.25 \text{ or } 25\% \end{aligned}$$

For Further Reference:

(Study Session 13, Modules 41.2, 41.3, LOS 41.g, 41.j)

SchweserNotes, Book 4 page 291

CFA® Program Curriculum, Volume 4, page 459

SchweserNotes, Book 4 page 297

CFA® Program Curriculum, Volume 4, page 468

Question #80 of 120

Question ID: 1207694

Which of the following statements about short sales is *least accurate*?

A) Proceeds from short sales cannot be withdrawn from the account.

B) The short seller must pay the lender of the stock any dividends paid by the company.

C) The short seller is required to replace the borrowed securities within six months of a short sale.

Explanation

There is no maximum time for which a security can be borrowed. It must be returned whenever the lender requires it to be.

For Further Reference:

(Study Session 12, Module 36.2, LOS 36.e)

SchweserNotes, Book 4 page 198

CFA® Program Curriculum, Volume 4, page 240

Question #81 of 120

Question ID: 1207701

Which of the following firms' earnings are likely to exhibit the greatest degree of sensitivity to the business cycle?

A) Furniture producer with high fixed costs as a proportion of total costs.

B) Entertainment producer with high variable costs as a proportion of total costs.

C) Food and beverage producer with high fixed costs as a proportion of total costs.

Explanation

Consumers buy fewer durable goods, such as furniture, during recessions and buy more during expansions. As a result, producers of these goods tend to have cyclical demand, revenues, and earnings. Operating leverage (high fixed costs as a proportion of total costs) also contributes to cyclicality of earnings.

For Further Reference:

(Study Session 13, Module 40.1, LOS 40.c)

SchweserNotes, Book 4 page 266

CFA® Program Curriculum, Volume 4, page 390

Question #82 of 120

Question ID: 1207700

The type of share voting *most likely* to result in significant minority shareholders having an approximately proportional representation on the board of directors is:

A) statutory voting.

B) weighted voting.

C) cumulative voting.

Explanation

Under cumulative voting, shareholders receive one vote per share for each board position election but can vote them for any board candidate or spread them over multiple candidates. This allows, for example, a holder of 20% of the shares to elect one of five board members. Under statutory voting, a minority shareholder could not elect any board members based only on their share voting rights.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.e and Study Session 13, Module 39.1, LOS 39.b)

SchweserNotes, Book 4 page 4

CFA® Program Curriculum, Volume 4, page 15

SchweserNotes, Book 4 page 251

CFA® Program Curriculum, Volume 4, page 358

Question #83 of 120

Question ID: 1207697

Which of the following indexes is *most likely* to be rebalanced on a regular basis?

A) Price-weighted index.
B) **Equal-weighted index.**
C) Market-capitalization weighted index.

Explanation

Equal-weighted indexes require regular and frequent rebalancing, because price changes of their component securities will cause component weights to drift away from their target weights. Price-weighted indexes and market-capitalization weighted indexes generally do not require regular rebalancing, because both the target weight and actual weight of each security varies with the price of that security.

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.f)

SchweserNotes, Book 4 page 225

CFA® Program Curriculum, Volume 4, page 293

Question #84 of 120

Question ID: 1207703

Over the most recent period, Ladden Materials has seen slow growth, increased competition, and declining profitability in its industry. The phase of the industry life cycle for Ladden's industry is *most likely*:

A) mature.
B) decline.
C) **shakeout.**

Explanation

The shakeout phase of the industry life cycle is characterized by slowing growth, intense competition, and declining profitability. The mature phase is characterized by industry consolidation and little or no growth. In the decline phase of the industry lifecycle, growth is negative and excess capacity results.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.h)

SchweserNotes, Book 4 page 272

CFA® Program Curriculum, Volume 4, page 414

Question #85 of 120

Question ID: 1207698

Mike Bowers observes that during one year the return on the S&P 500 index is 20%. Recalculating the return on an equally weighted basis, Bowers estimates that the index return is 15%. The difference in the two calculations of return is *best* explained by:

A) **large capitalization stocks outperforming small capitalization stocks.**
B) small capitalization stocks outperforming large capitalization stocks.
C) dividends on the stocks in the index.

Explanation

Because the S&P 500 index is market capitalization weighted, stocks with higher market capitalization have greater influence on the performance of the index. Because the index outperformed its equally weighted version, larger capitalization stocks performed better than smaller capitalization stocks.

For Further Reference:

(Study Session 12, Modules 37.1, 37.2, LOS 37.d, 37.e, 37.k)

CFA® Program Curriculum, Volume 4, page 285

CFA® Program Curriculum, Volume 4, page 285

CFA® Program Curriculum, Volume 4, page 305

Question #86 of 120

Question ID: 1207696

Which of the following is *most accurate* regarding the relationship between operational efficiency and informational efficiency?

A) **Operational efficiency contributes to informational efficiency.**
B) Informational efficiency is independent of operational efficiency.
C) There is a trade-off between operational efficiency and informational efficiency.

Explanation

An operationally efficient market is characterized by low trading costs. The low trading costs of an operationally efficient market will encourage trading based on new information, thus making security prices more informationally efficient. High trading costs limit traders' ability to exploit violations of informational efficiency.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.k)

CFA® Program Curriculum, Volume 4, page 260

Question #87 of 120

Question ID: 1207714

Questions 87 through 99 relate to Fixed Income. (19.5 minutes)

Compared to a bond's Macaulay duration, its modified duration:

A) **is lower.**

B) is higher.

C) may be lower or higher.

Explanation

Modified duration = Macaulay duration / (1 + YTM). Modified duration is lower than Macaulay duration unless YTM equals zero.

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.b)

SchweserNotes, Book 5 page 103

CFA® Program Curriculum, Volume 5, page 248

Question #88 of 120

Question ID: 1207711

An analyst obtains a market quote for the two-year forward rate two years from now. To derive the next point on a theoretical annual forward rate curve, the analyst can use:

A) the two-year and five-year spot rates.

B) the three-year and four-year spot rates.

C) **the three-year and five-year spot rates.**

Explanation

Given the two-year forward rate two years from now, the next point on an annual forward rate curve is the two-year forward rate three years from now, 3y2y. This rate can be derived from the three-year and five-year spot rates as follows: $(1 + S_5)^5 = (1 + S_3)^3 (1 + 3y2y)^2$.

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.i, 44.j)

SchweserNotes, Book 5 page 58

CFA® Program Curriculum, Volume 5, page 140

SchweserNotes, Book 5 page 58

CFA® Program Curriculum, Volume 5, page 143

Question #89 of 120

Question ID: 1207719

The credit rating agency practice of "notching" is *best* described as:

A) **assigning different ratings to different debt issues from the same issuer.**

B) downgrading or upgrading the rating of a debt issue or issuer by one increment.

C) adding a plus or minus sign to a rating to indicate a positive or negative outlook.

Explanation

"Notching" refers to the credit rating agency practice of assigning ratings to debt issues that differ from the issuer's credit rating. An issuer credit rating applies to a firm's senior unsecured debt. Debt issues with different seniority or covenants may be notched to a higher or lower issue credit rating.

For Further Reference:

(Study Session 15, Module 47.1, LOS 47.d)

SchweserNotes, Book 5 page 129

CFA® Program Curriculum, Volume 5, page 313

Question #90 of 120

Question ID: 1207717

A bond priced at par ($1,000) has a modified duration of 8 and a convexity of 100. If interest rates fall 50 basis points, the new price will be *closest* to:

A) $1,041.25.

B) $958.75.

C) $875.00.

Explanation

$$\frac{\Delta P}{P} = -\text{Duration}(\Delta \text{YTM}) + \frac{1}{2}\text{Convexity}(\Delta \text{YTM})^2$$

$$\begin{aligned}\frac{\Delta P}{P} &= (-)(8)(-0.005) + \frac{1}{2}(100)(-0.005)^2 \\ &= +0.0400 + 0.00125 \\ &= +0.04125\text{, or up } 4.125\%\end{aligned}$$

The price would thus be $1,000 × 1.04125 = $1,041.25.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.i)

SchweserNotes, Book 5 page 114

CFA® Program Curriculum, Volume 5, page 270

Question #91 of 120

Question ID: 1207710

An analyst needs to estimate the value of an illiquid 7% BB+ rated bond that has eight years to maturity. Using matrix pricing, the analyst should *most appropriately* base an estimate for this bond on yields of:

A) on-the-run eight-year government bonds.

B) more frequently traded bonds rated BB+.

C) other BB+ rated bonds with similar liquidity to this bond.

Explanation

Matrix pricing for untraded or infrequently traded bonds should be based on yields of more frequently traded bonds with similar credit ratings.

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.e)

SchweserNotes, Book 5 page 43

CFA® Program Curriculum, Volume 5, page 123

Question #92 of 120

Question ID: 1207720

Acme Holdings operates in an industry for which three-year average financial ratios by credit rating are as follows:

Ratio	AAA	AA	A	BBB	BB	B	CCC
FCF/debt	32.0%	25.9%	21.8%	18.7%	12.3%	7.0%	3.1%
Debt/EBITDA	0.9×	1.3×	1.5×	1.9×	2.3×	3.5×	5.0×

If Acme has a three-year average debt-to-EBITDA ratio of 2.4 and a free cash flow to debt ratio of 7.1, its credit rating is *most likely* to be:

A) investment grade.
B) **below investment grade.**
C) borderline investment grade.

Explanation

Acme's financial ratios suggest its credit rating should be between BB and B, which is below investment grade.

For Further Reference:

(Study Session 15, Module 47.2, LOS 47.h)

SchweserNotes, Book 5 page 137

CFA® Program Curriculum, Volume 5, page 618

Question #93 of 120

Question ID: 1207713

Commercial mortgage-backed securities (CMBS) loans typically have greater call protection than agency MBS loans because:

A) **commercial mortgages may have yield maintenance charges.**
B) smaller-sized mortgages typically are not refinanced if interest rates fall.
C) CMBS typically receive higher credit ratings from credit agencies than residential MBS.

Explanation

Any type of call protection structured into the loan itself (in this case, yield maintenance charges) increases the overall call protection of the CMBS. Agency MBS do not provide call protection at the individual loan level.

For Further Reference:

(Study Session 14, Module 45.2, LOS 45.g)

SchweserNotes, Book 5 page 87

CFA® Program Curriculum, Volume 5, page 208

Question #94 of 120

Question ID: 1207709

A non-amortizing fixed income security is *most accurately* described as:

- **A) a bullet bond.**
- **B)** a balloon bond.
- **C)** a mortgage bond.

Explanation

Bonds with a bullet structure are non-amortizing and return their entire principal to the bondholder at the maturity date. A non-amortizing bond makes a bullet *payment* at maturity.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.e)

SchweserNotes, Book 5 page 7

CFA® Program Curriculum, Volume 5, page 26

Question #95 of 120

Question ID: 1207708

For a domestic investor purchasing bonds in a foreign market and currency:

- **A) appreciation of both the asset and the foreign currency benefits the domestic investor.**
- **B)** depreciation of both the asset and the foreign currency benefits the domestic investor.
- **C)** appreciation of the asset and depreciation of the foreign currency benefit the domestic investor.

Explanation

When the foreign currency appreciates, each foreign currency-denominated cash flow buys more domestic currency units—increasing the domestic currency return from the investment. The appreciation of the foreign asset benefits the investor as well.

For Further Reference:

(Study Session 5, Module 18.1, LOS 18.c, Study Session 14, Module 42.1, LOS 42.a, 24.d)

SchweserNotes, Book 2 page 159

CFA® Program Curriculum, Volume 2, page 414

SchweserNotes, Book 5 page 1

CFA® Program Curriculum, Volume 5, page 7

SchweserNotes, Book 5 page 4

CFA® Program Curriculum, Volume 5, page 21

Question #96 of 120

Question ID: 1207715

Reinvestment risk is *least likely*:

A) minimized with zero-coupon bond issues.

B) more problematic for those investors with longer time horizons.

C) **more problematic when the current coupons being reinvested are relatively small.**

Explanation

Reinvestment risk becomes more problematic when the current coupons being reinvested are relatively large.

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.a)

SchweserNotes, Book 5 page 97

CFA® Program Curriculum, Volume 5, page 241

Question #97 of 120

Question ID: 1207718

A duration gap is *most accurately* described as a difference between a bond's:

A) Macaulay duration and effective duration.

B) **duration and the bondholder’s investment horizon.**

C) actual change in value and the change estimated using duration and convexity.

Explanation

"Duration gap" refers to a difference between a bond's Macaulay duration and the bondholder's investment horizon.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.k)

SchweserNotes, Book 5 page 115

CFA® Program Curriculum, Volume 5, page 279

Question #98 of 120

Question ID: 1207716

Siegel, Inc., has issued bonds maturing in 15 years but callable at any time after the first 8 years. The bonds have a coupon rate of 6%, and are currently trading at $992 per $1,000 par value. If interest rates decline over the next few years:

A) the call option embedded in the bonds will increase in value, but the price of the bond will decrease.

B) **the price of the bond will increase, but likely by less than a comparable bond with no embedded option.**

C) the price of the bond will increase, primarily as a result of the increasing value of the call option.

Explanation

The value of the bond would increase due to the lower interest rates, but the increasing value of the call option would offset some portion of this increase in bond value because the bondholder is short the call option.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.f, Study Session 15, Module 46.2 , LOS 46.e)

SchweserNotes, Book 5 page 11

CFA® Program Curriculum, Volume 5, page 37

SchweserNotes, Book 5 page 108

CFA® Program Curriculum, Volume 5, page 259

Question #99 of 120

Question ID: 1207712

Jefferson Blake, CFA, believes there is a good opportunity to purchase an option-free 4% annual pay bond with three years left until maturity, a zero-volatility spread of 40 basis points, and a par value of $1,000. Blake observes that 1-year, 2-year, and 3-year government bond spot rates are currently 4.0%, 4.5%, and 4.75%, respectively. The maximum price Blake should be willing to pay for the bond is *closest* to:

A) $940.
B) **$970.**
C) $980.

Explanation

To compute the value of this bond, discount each of the cash flows at a different interest rate appropriate for the timing of the cash flow. The appropriate rates are 40 bp greater than the spot rates.

$$\frac{40}{(1.0400+0.0040)}+\frac{40}{(1.0450+0.0040)^2}+\frac{(40+1{,}000)}{(1.0475+0.0040)^3}=969.22$$

For Further Reference:

(Study Session 14, Module 44.2, 44.5, LOS 44.c, 44.k)

CFA® Program Curriculum, Volume 5, page 117

CFA® Program Curriculum, Volume 5, page 147

Question #100 of 120

Question ID: 1207725

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

Which of the following is *most likely* to increase the price of a forward contract on an asset?

A) Higher dividends from a stock.

B) Lower storage costs for a commodity.

C) Lower convenience yield for a commodity.

Explanation

Either a decrease in benefits or an increase in costs of holding the underlying asset would increase the no-arbitrage price of a forward contract.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.e)

SchweserNotes, Book 5 page 174

CFA® Program Curriculum, Volume 5, page 471

Question #101 of 120

Question ID: 1207723

A call option on a $25 stock with an exercise price of $30 sells for a premium of $4. At expiration, the writer of the call will experience a net loss on the option if the stock price is greater than:

A) $26.

B) $29.

C) $34.

Explanation

The call writer will experience a net loss on the call option if the stock's price exceeds X + C = $30 + $4 = $34.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.d)

SchweserNotes, Book 5 page 164

CFA® Program Curriculum, Volume 5, page 406

Question #102 of 120

Question ID: 1207727

A bank borrows for 360 days and simultaneously lends the proceeds for 90 days. This transaction creates a synthetic forward rate agreement (FRA) *closest* to:

A) a long position in a 90-day FRA on 270-day LIBOR.

B) a long position in a 90-day FRA on 360-day LIBOR.

C) a short position in a 360-day FRA on 90-day LIBOR.

Explanation

If a bank borrows for 360 days and simultaneously lends the proceeds for 90 days, it creates a synthetic long (borrower) position in a 90-day FRA on 270-day LIBOR. The bank has no net position for the first 90 days and a borrowing position at a fixed rate of interest for the subsequent 270 days.

For Further Reference:

(Study Session 16, Module 49.2, LOS 49.f)

SchweserNotes, Book 5 page 174

CFA® Program Curriculum, Volume 5, page 473

Question #103 of 120

Question ID: 1207728

An American-style call option is *most likely* to be more valuable than an otherwise equivalent European-style call option if:

A) the call is deep in the money.

B) **its underlying asset is a semiannual-pay bond.**

C) implied volatility increases during the life of the call.

Explanation

American options can be more valuable than otherwise equivalent European options only if early exercise has value to the option holder. For call options, early exercise may be valuable if the underlying asset provides cash flows. Early exercise may be valuable for put options, particularly if they are deep in the money.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.o)

SchweserNotes, Book 5 page 187

CFA® Program Curriculum, Volume 5, page 500

Question #104 of 120

Question ID: 1207722

Which of the following statements about futures and forwards is *most accurate*? Futures:

A) are subject to default risk, but forwards are not.

B) are individualized contracts, but forwards are standardized.

C) **require that traders post margin in order to trade, but forwards typically require no cash transaction until the delivery date.**

Explanation

Forwards are subject to default risk, but *futures* are not. *Forwards* are individualized contracts, but *futures* are standardized.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.b)

SchweserNotes, Book 5 page 158

CFA® Program Curriculum, Volume 5, page 387

Question #105 of 120

Question ID: 1207724

Derivatives markets are *most likely* to:

A) **reduce transactions costs.**

B) increase speculation and risk.

C) provide arbitrage opportunities to investors.

Explanation

The key advantages of derivatives markets are providing price information, reducing transactions costs, and shifting risks among market participants. Derivatives markets are highly efficient and arbitrage opportunities rarely exist or are quickly eliminated.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.e)

SchweserNotes, Book 5 page 164

CFA® Program Curriculum, Volume 5, page 422

Question #106 of 120

Question ID: 1207726

Derivatives pricing is based on the assumption that:

A) no arbitrage occurs.

B) **the law of one price holds.**

C) long and short investors are net risk-neutral.

Explanation

Derivatives pricing is based on no-arbitrage pricing and replication, and therefore assumes the law of one price holds (i.e., two assets or portfolios with the same future payoffs must have the same price). Derivatives pricing does not assume long and short investors are net risk-neutral; rather, it determines no-arbitrage derivative prices using risk-neutral pricing.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.a)

CFA® Program Curriculum, Volume 5, page 456

Question #107 of 120

Question ID: 1207732

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

A commodities market is typically in backwardation if:

A) speculators are the primary drivers of futures prices.

B) users of a commodity are the primary drivers of futures prices.

C) **producers of a commodity are the primary drivers of futures prices.**

Explanation

When commodity producers sell futures to hedge their price risk, futures prices are driven down compared to spot prices, resulting in backwardation.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 145

Question #108 of 120

Question ID: 1207729

To exit an investment in a portfolio company through a trade sale, a private equity firm sells:

A) shares of a portfolio company to the public.
B) the portfolio company to another private equity firm.
C) **the portfolio company to one of the portfolio company's competitors.**

Explanation

A trade sale involves selling a portfolio company to a competitor or another strategic buyer. An IPO involves selling all or some shares of a portfolio company to the public. A secondary sale involves selling a portfolio company to another private equity firm or a group of investors.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

SchweserNotes, Book 5 page 199

CFA® Program Curriculum, Volume 6, page 10

Question #109 of 120

Question ID: 1207733

Which of the following is *least likely* a benefit of fund of funds (FOF) investing?

A) FOFs may permit access to otherwise unavailable hedge funds.
B) FOFs allow investors to diversify the risks of holding a single hedge fund.
C) **The fee is generally quite reasonable since the investor only pays the manager of the FOF.**

Explanation

The fee may actually be substantial since, in addition to paying the manager of the FOF, a fee must be paid to each hedge fund within the FOF.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

SchweserNotes, Book 5 page 199

CFA® Program Curriculum, Volume 6, page 10

Question #110 of 120

Question ID: 1207730

A private equity firm that provides equity capital to a publicly traded company to finance the company's restructuring, but does not take the company private, is *best* described as engaging in:

A) angel investing.
B) mezzanine financing.
C) **private investment in public equity.**

Explanation

Private investment in public equities refers to a private equity firm providing equity financing to publicly traded companies. Angel investing refers to financing the formation of a business. Mezzanine financing refers to subordinated equity-linked debt or preferred shares issued to finance a leveraged buyout.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

SchweserNotes, Book 5 page 199

CFA® Program Curriculum, Volume 6, page 10

Question #111 of 120

Question ID: 1207734

Which provision of a hedge fund's incentive fees is designed to prevent investors from paying multiple incentive fees for the same performance?

A) Hurdle rate.
B) **High water mark.**
C) 2-and-20 structure.

Explanation

If a hedge fund has a high water mark, incentive fees are based on increases in investors' accounts above their highest previous values. As a result, fund managers do not receive incentive fees on returns that reverse previous losses. A hurdle rate is a minimum return a fund must achieve in a given period before fund managers receive incentive fees. A 2-and-20 structure means fund managers receive a 2% management fee and a 20% incentive fee; other provisions determine the base amounts on which these fees are calculated.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

SchweserNotes, Book 5 page 209

CFA® Program Curriculum, Volume 6, page 20

Question #112 of 120

Question ID: 1207731

Which of the following is *most likely* to represent the management fees of a private equity fund?

A) 2% of invested capital.
B) **3% of committed capital.**
C) 1% of drawn-down capital.

Explanation

Private equity management fees are typically 1% to 3% of committed capital rather than invested (drawn-down) capital.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

SchweserNotes, Book 5 page 199

CFA® Program Curriculum, Volume 6, page 10

Question #113 of 120

Question ID: 1207735

A hedge fund's lockup period is the time during which:

A) the fund is closed to new investors.

B) an investor may not redeem shares in the fund.

C) the fund must return cash to an investor who has requested a redemption.

Explanation

During a lockup period, an investor is restricted from redeeming shares.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

SchweserNotes, Book 5 page 199

CFA® Program Curriculum, Volume 6, page 10

Question #114 of 120

Question ID: 1207736

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

Which of the following portfolios will have the lowest diversification ratio? A portfolio of:

A) 30 equally-weighted stocks with companies from the same industry.

B) 20 equally-weighted stocks with companies from different industries.

C) 30 equally-weighted stocks with companies from different industries.

Explanation

The diversification ratio of a portfolio equals its standard deviation of returns divided by the average standard deviation of the individual securities in the portfolio. Therefore, a more diversified portfolio will have a lower diversification ratio than a less diversified portfolio. A portfolio containing the highest number of securities from different industries will be the most diversified and will have the lowest diversification ratio. A portfolio of stocks from the same industry is likely to have a higher diversification ratio (reflecting less diversification) than a portfolio of stocks from different industries.

For Further Reference:

(Study Session 18, Module 51.1, LOS 51.a)

SchweserNotes, Book 4 page 102

CFA® Program Curriculum, Volume 6, page 80

Question #115 of 120

Question ID: 1207741

In a case where a client's capacity to bear risk is significantly less than the client's expressed willingness to bear risk, the *most appropriate* action for a financial advisor is to:

A) counsel the client and attempt to change his attitude towards risk.

B) **base the assessment of risk tolerance in the IPS on client's ability to bear risk.**

C) attempt to educate the client about investment risk and correct any misconceptions.

Explanation

In a situation where the client's expressed willingness to bear investment risk is significantly greater than the client's ability to bear investment risk, the advisor's assessment of the client's risk tolerance in the IPS should reflect the client's ability to bear investment risk.

For Further Reference:

(Study Session 19, Module 54.1, LOS 54.d)

SchweserNotes, Book 4 page 176

CFA® Program Curriculum, Volume 6, page 252

Question #116 of 120

Question ID: 1207740

Which of the following statements about risk is *most accurate*?

A) The capital market line plots expected return against market risk.

B) The efficient frontier plots expected return against unsystematic risk.

C) **The security market line plots expected return against systematic risk.**

Explanation

The capital market line plots expected return against standard deviation of returns for efficient portfolios. The efficient frontier plots expected return against the standard deviation of return, a measure of total risk.

For Further Reference:

(Study Session 18, Modules 52.3, LOS 52.g, Module 53.1, 53.2, LOS 53.b, 53.f)

SchweserNotes, Book 4 page 136

CFA® Program Curriculum, Volume 6, page 163

SchweserNotes, Book 4 page 149

CFA® Program Curriculum, Volume 6, page 192

SchweserNotes, Book 4 page 158

CFA® Program Curriculum, Volume 6, page 213

Question #117 of 120

Question ID: 1207739

A portfolio manager is constructing a new equity portfolio consisting of a large number of randomly chosen domestic stocks. As the number of stocks in the portfolio increases, what happens to the expected levels of systematic and unsystematic risk?

	Systematic risk	Unsystematic risk
A)	Increases	Remains the same
B)	Decreases	Increases
C)	**Remains the same**	**Decreases**

Explanation

As randomly selected securities are added to a portfolio, the diversifiable (unsystematic) risk decreases, and the expected level of nondiversifiable (systematic) risk remains the same.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.c)

SchweserNotes, Book 4 page 153

CFA® Program Curriculum, Volume 6, page 204

Question #118 of 120

Question ID: 1207742

The type of technical analysis chart *most likely* to be useful for intermarket analysis is a:

- **A)** candlestick chart.
- **B)** point-and-figure chart.
- **C)** **relative strength chart.**

Explanation

Relative strength charts display the price of an asset relative to the price of another asset or benchmark over time. This type of chart is useful for demonstrating whether one asset class or market has outperformed or underperformed another. Candlestick charts and point-and-figure charts are generally used to display price patterns for a single asset or market over time.

For Further Reference:

(Study Session 19, Module 56.1, LOS 56.b, 56.h)

SchweserNotes, Book 1 page 315

CFA® Program Curriculum, Volume 6, page 342

SchweserNotes, Book 1 page 327

CFA® Program Curriculum, Volume 6, page 388

Question #119 of 120

Question ID: 1207738

Rolly Parker, CFA, has managed the retirement account funds for Misto Inc. for the last two years. Contributions and withdrawals from the account are decided by Misto's CFO. The account history is as follows, with account values calculated before same-date deposits and withdrawals:

Jan 1, 20X1	Beginning portfolio value	$10 million
Jul 1, 20X1	Account value	$11.2 million
Jul 1, 20X1	Deposit of cash	$1.2 million
Jan 1, 20X2	Account value	$12.5 million
Jan 1, 20X2	Withdrawal of cash	$0.6 million
Dec 31, 20X2	Account value	$15 million

The appropriate annual return to use in evaluating the manager's performance is *closest* to:

A) 9%.
B) **19%.**
C) 22%.

Explanation

Since the portfolio manager is not directing the flow of cash into and out of the account, the time-weighted annual rate of return is the appropriate performance measure. Calculate the 2-year holding period return +1, then take the square root and subtract 1 to get the annual time-weighted rate of return: $[(11.2 / 10)(12.5 / 12.4)(15 / 11.9)]^{1/2} - 1 = 19.29\%$.

For Further Reference:

(Study Session 18, Module 52.1, LOS 52.a)

SchweserNotes, Book 4 page 125

CFA® Program Curriculum, Volume 6, page 116

Question #120 of 120

Question ID: 1207737

Which of the following pooled investment shares is *least likely* to trade at a price different from its NAV?

A) Exchange-traded fund shares.
B) **Open-end mutual fund shares.**
C) Closed-end mutual fund shares.

Explanation

Shares of open-end mutual funds trade at NAV. The others may deviate from NAV.

For Further Reference:

(Study Session 18, Module 51.2, LOS 51.f)

SchweserNotes, Book 4 page 105

CFA® Program Curriculum, Volume 6, page 104